

SAHAYI Platform - Database Schema

Comprehensive Relational Database Data Dictionary & Table Structure

1. tbl_UserRoles

Table_Name :	tbl_UserRoles
Primary Key :	RoleId
Foreign Key :	None

Description: Stores system roles for Role-Based Access Control (RBAC) to enforce security and access segregation.

Column Name	Data Type	Constraints	Description
RoleId	INT	PRIMARY KEY, IDENTITY(1,1)	Unique auto-incrementing role identifier
RoleName	NVARCHAR(50)	NOT NULL, UNIQUE	System role name (e.g., CDS_Admin, President, Secretary, Treasurer, Member)
Description	NVARCHAR(255)	NULL	Detailed summary of role responsibilities and permission scope

2. tbl_PanchayathWards

Table_Name :	tbl_PanchayathWards
Primary Key :	WardId
Foreign Key :	None

Description: Defines official ward administrative boundaries under the local Panchayath governance structure.

Column Name	Data Type	Constraints	Description
WardId	INT	PRIMARY KEY, IDENTITY(1,1)	Unique Ward Identifier
WardNumber	INT	NOT NULL, UNIQUE	Official assigned ward number within Panchayath
WardName	NVARCHAR(100)	NOT NULL	Local ward/area locality name

3. tbl_AyalkoottamUnits

Table_Name :	tbl_AyalkoottamUnits
Primary Key :	UnitId

Foreign Key :	WardId -> tbl_PanchayathWards(WardId)		
Description: Stores registered Kudumbashree Ayalkoottam unit profiles created and onboarded by CDS personnel.			
Column Name	Data Type	Constraints	Description
UnitId	UNIQUEIDENTIFIER	PRIMARY KEY, DEFAULT NEWID()	Unique Globally Unique Identifier (GUID) for the unit
UnitName	NVARCHAR(150)	NOT NULL	Official name of the Ayalkoottam Self-Help Group unit
WardId	INT	FOREIGN KEY	Associated ward location ID (Links to tbl_PanchayathWards)
AccountNumber	NVARCHAR(50)	NOT NULL	Bank account number of the Ayalkoottam unit
BankName	NVARCHAR(100)	NOT NULL	Bank name where the unit account is maintained
IFSCCode	NVARCHAR(20)	NOT NULL	IFSC Code of the bank branch
CreatedDate	DATETIME2	DEFAULT GETUTCDATE(), NOT NULL	Unit onboarding timestamp
IsActive	BIT	DEFAULT 1, NOT NULL	Active operational status flag (1 = Active, 0 = Inactive)

4. tbl_ApplicationUsers

Table_Name :	tbl_ApplicationUsers		
Primary Key :	UserId		
Foreign Key :	RoleId -> tbl_UserRoles(RoleId), UnitId -> tbl_AyalkoottamUnits(UnitId)		
Description: Stores profile details, contact information, and authentication credentials for all platform users.			
Column Name	Data Type	Constraints	Description
UserId	UNIQUEIDENTIFIER	PRIMARY KEY, DEFAULT NEWID()	Unique GUID identifying the user
FullName	NVARCHAR(150)	NOT NULL	Full legal name of the user
PhoneNumber	NVARCHAR(15)	NOT NULL, UNIQUE	Mobile phone number used as primary credential for login
HouseName	NVARCHAR(150)	NOT NULL	Residential address house name / details
PasswordHash	NVARCHAR(MAX)	NOT NULL	BCrypt encrypted password hash
IsPasswordChanged	BIT	DEFAULT 0, NOT NULL	Tracks initial login status to force

Column Name	Data Type	Constraints	Description
			initial password update
RoleId	INT	FOREIGN KEY	Assigned access role ID (Links to tbl_UserRoles)
UnitId	UNIQUEIDENTIFIER	FOREIGN KEY (NULLABLE)	Associated Ayalkoottam unit ID (NULL for CDS Admins)
JoinedDate	DATETIME2	DEFAULT GETUTCDATE(), NOT NULL	Account registration timestamp
IsActive	BIT	DEFAULT 1, NOT NULL	User account status (1 = Active, 0 = Deactivated)

5. tbl_Meetings

Table_Name :	tbl_Meetings
Primary Key :	MeetingId
Foreign Key :	UnitId -> tbl_AyalkoottamUnits(UnitId), CreatedBy -> tbl_ApplicationUsers(UserId)

Description: Logs scheduled weekly group governance meetings, venue locations, and recorded Minutes of Meeting (MoM).

Column Name	Data Type	Constraints	Description
MeetingId	INT	PRIMARY KEY, IDENTITY(1,1)	Unique Meeting Record Identifier
UnitId	UNIQUEIDENTIFIER	FOREIGN KEY	Target Ayalkoottam unit ID
MeetingDate	DATETIME2	NOT NULL	Scheduled meeting date and time
Venue	NVARCHAR(200)	NOT NULL	Meeting physical location / host member venue
MinutesOfMeeting	NVARCHAR(MAX)	NULL	Detailed meeting minutes recorded by Secretary
CreatedBy	UNIQUEIDENTIFIER	FOREIGN KEY	Secretary User ID who scheduled and recorded meeting

6. tbl_Attendances

Table_Name :	tbl_Attendances
Primary Key :	AttendanceId

Foreign Key : MeetingId -> tbl_Meetings(MeetingId), UserId -> tbl_ApplicationUsers(UserId)

Description: Tracks individual member attendance and presence records for each scheduled weekly meeting.

Column Name	Data Type	Constraints	Description
AttendanceId	INT	PRIMARY KEY, IDENTITY(1,1)	Unique Attendance Record Identifier
MeetingId	INT	FOREIGN KEY	Target meeting ID (Links to tbl_Meetings)
UserId	UNIQUEIDENTIFIER	FOREIGN KEY	Attending member User ID (Links to tbl_ApplicationUsers)
IsPresent	BIT	NOT NULL	Attendance status (1 = Present, 0 = Absent)
MarkedAt	DATETIME2	DEFAULT GETUTCDATE(), NOT NULL	Attendance recording timestamp

7. tbl_SavingsTransactions

Table_Name :	tbl_SavingsTransactions
Primary Key :	TransactionId
Foreign Key :	UserId -> tbl_ApplicationUsers(UserId), UnitId -> tbl_AyalkoottamUnits(UnitId), RecordedBy -> tbl_ApplicationUsers(UserId)

Description: Digital passbook ledger recording weekly thrift savings deposits collected from members during meetings.

Column Name	Data Type	Constraints	Description
TransactionId	INT	PRIMARY KEY, IDENTITY(1,1)	Unique Savings Ledger Entry ID
UserId	UNIQUEIDENTIFIER	FOREIGN KEY	Depositing member User ID
UnitId	UNIQUEIDENTIFIER	FOREIGN KEY	Member's Ayalkoottam unit ID
Amount	DECIMAL(18,2)	NOT NULL, CHECK (Amount > 0)	Weekly thrift savings amount deposited
TransactionDate	DATETIME2	DEFAULT GETUTCDATE(), NOT NULL	Deposit transaction date
ReceiptNumber	NVARCHAR(50)	NOT NULL, UNIQUE	Auto-generated digital transaction receipt number
RecordedBy	UNIQUEIDENTIFIER	FOREIGN KEY	Treasurer/Secretary User ID who

Column Name	Data Type	Constraints	Description
	IER		collected deposit

8. tbl_LoanApplications

Table_Name :	tbl_LoanApplications
Primary Key :	LoanId
Foreign Key :	UserId -> tbl_ApplicationUsers(UserId), UnitId -> tbl_AyalkoottamUnits(UnitId), ApprovedBy -> tbl_ApplicationUsers(UserId)

Description: *Manages micro-loan requests, eligibility reviews, approval workflows, and loan lifecycle terms.*

Column Name	Data Type	Constraints	Description
LoanId	INT	PRIMARY KEY, IDENTITY(1,1)	Unique Loan Application Identifier
UserId	UNIQUEIDENTIFIER	FOREIGN KEY	Borrower member User ID
UnitId	UNIQUEIDENTIFIER	FOREIGN KEY	Associated Ayalkoottam unit ID
AmountRequested	DECIMAL(18,2)	NOT NULL, CHECK (> 0)	Requested loan principal amount
Purpose	NVARCHAR(500)	NOT NULL	Stated justification and purpose for loan
TenureMonths	INT	NOT NULL, CHECK (> 0)	Repayment period duration in months
InterestRate	DECIMAL(5,2)	NOT NULL, CHECK (>= 0)	Annual reducing interest rate percentage
Status	NVARCHAR(20)	DEFAULT 'Pending', NOT NULL	Application state (Pending, Approved, Rejected, Disbursed, Closed)
AppliedDate	DATETIME2	DEFAULT GETUTCDATE(), NOT NULL	Application submission timestamp
ApprovedBy	UNIQUEIDENTIFIER	FOREIGN KEY (NULLABLE)	Secretary/President User ID who reviewed and approved
DisbursedDate	DATETIME2	NULL	Fund disbursement timestamp by Treasurer

9. tbl_LoanRepayments

Table_Name :	tbl_LoanRepayments
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Primary Key :	RepaymentId
Foreign Key :	LoanId -> tbl_LoanApplications(LoanId), RecordedBy -> tbl_ApplicationUsers(UserId)

Description: Tracks monthly EMI repayment installments, splitting principal reduction and reducing interest components.

Column Name	Data Type	Constraints	Description
RepaymentId	INT	PRIMARY KEY, IDENTITY(1,1)	Unique Repayment Entry Identifier
LoanId	INT	FOREIGN KEY	Target micro-loan ID (Links to tbl_LoanApplications)
AmountPaid	DECIMAL(18,2)	NOT NULL, CHECK (> 0)	Total EMI amount paid
PrincipalComponent	DECIMAL(18,2)	NOT NULL, CHECK (>= 0)	Portion allocated to reduce loan principal balance
InterestComponent	DECIMAL(18,2)	NOT NULL, CHECK (>= 0)	Portion calculated for reducing interest charge
RepaymentDate	DATETIME2	DEFAULT GETUTCDATE(), NOT NULL	Repayment payment timestamp
ReceiptNumber	NVARCHAR(50)	NOT NULL, UNIQUE	Unique payment receipt reference number
RecordedBy	UNIQUEIDENTIFIER	FOREIGN KEY	Treasurer User ID who collected payment

10. tbl_Notifications

Table_Name :	tbl_Notifications
Primary Key :	NotificationId
Foreign Key :	UserId -> tbl_ApplicationUsers(UserId)

Description: Stores system alerts, meeting schedules, thrift payment due notices, and loan update reminders.

Column Name	Data Type	Constraints	Description
NotificationId	INT	PRIMARY KEY, IDENTITY(1,1)	Unique Notification Identifier
UserId	UNIQUEIDENTIFIER	FOREIGN KEY	Target recipient User ID
Title	NVARCHAR(100)	NOT NULL	Notification header / subject title
Message	NVARCHAR(500)	NOT NULL	Detailed alert or reminder body text

Column Name	Data Type	Constraints	Description
IsRead	BIT	DEFAULT 0, NOT NULL	Read status flag (0 = Unread, 1 = Read)
CreatedDate	DATETIME2	DEFAULT GETUTCDATE(), NOT NULL	Notification dispatch timestamp

11. tbl_ChatGroups

Table_Name :	tbl_ChatGroups
Primary Key :	GroupId
Foreign Key :	UnitId -> tbl_AyalkoottamUnits(UnitId)

Description: Stores unit-wide group chat room channels automatically provisioned for Ayalkoottam units.

Column Name	Data Type	Constraints	Description
GroupId	INT	PRIMARY KEY, IDENTITY(1,1)	Unique Group Chat ID
UnitId	UNIQUEIDENTIFIER	FOREIGN KEY	Associated Ayalkoottam unit ID
GroupName	NVARCHAR(100)	NOT NULL	Group title (e.g., 'Ward 5 - Jasmine Unit Group')
CreatedDate	DATETIME2	DEFAULT GETUTCDATE(), NOT NULL	Group channel creation timestamp

12. tbl_GroupMessages

Table_Name :	tbl_GroupMessages
Primary Key :	GroupMessageId
Foreign Key :	GroupId -> tbl_ChatGroups(GroupId), SenderId -> tbl_ApplicationUsers(UserId)

Description: Stores text messages exchanged inside unit-wide group chat channels via SignalR.

Column Name	Data Type	Constraints	Description
GroupMessageId	BIGINT	PRIMARY KEY, IDENTITY(1,1)	Unique Group Message ID
GroupId	INT	FOREIGN KEY	Target Group Chat ID
SenderId	UNIQUEIDENTIFIER	FOREIGN KEY	User ID of the member sending the message

Column Name	Data Type	Constraints	Description
MessageText	NVARCHAR(MAX)	NOT NULL	Text content of the message
SentAt	DATETIME2	DEFAULT GETUTCDATE(), NOT NULL	Message dispatch timestamp
IsDeleted	BIT	DEFAULT 0, NOT NULL	Soft deletion indicator flag

13. tbl_DirectMessages

Table_Name :	tbl_DirectMessages
Primary Key :	DirectMessageId
Foreign Key :	SenderId -> tbl_ApplicationUsers(UserId), ReceiverId -> tbl_ApplicationUsers(UserId)

Description: Stores private one-on-one messages between members or office bearers within the platform.

Column Name	Data Type	Constraints	Description
DirectMessageId	BIGINT	PRIMARY KEY, IDENTITY(1,1)	Unique Direct Message ID
SenderId	UNIQUEIDENTIFIER	FOREIGN KEY	Sender User ID (Links to tbl_ApplicationUsers)
ReceiverId	UNIQUEIDENTIFIER	FOREIGN KEY	Recipient User ID (Links to tbl_ApplicationUsers)
MessageText	NVARCHAR(MAX)	NOT NULL	Private message content text
SentAt	DATETIME2	DEFAULT GETUTCDATE(), NOT NULL	Message dispatch timestamp
IsRead	BIT	DEFAULT 0, NOT NULL	Read receipt status flag (0 = Unread, 1 = Read)